

Task3: Collections & Exception Handling

Code:

```
import java.util.*;

class InvalidAgeException extends Exception {

    public InvalidAgeException(String message) {

        super(message);

    }

}

public class CollectionsDemo {

    public static void checkAge(int age) throws InvalidAgeException {

        if (age < 18) {

            throw new InvalidAgeException("Age must be 18 or above.");

        }

        System.out.println("Valid Age");

    }

    public static void main(String[] args) {

        // ArrayList Example

        ArrayList<String> students = new ArrayList<>();

        students.add("Isha");

        students.add("Rahul");

        students.add("Aman");

        System.out.println("ArrayList:");

        for (String student : students) {

            System.out.println(student);

        }

    }

}
```

```
}
```

```
// HashMap
```

```
HashMap<Integer, String> studentMap = new HashMap<>();
```

```
studentMap.put(101, "Isha");
```

```
studentMap.put(102, "Rahul");
```

```
System.out.println("\nHashMap:");
```

```
for (Map.Entry<Integer, String> entry : studentMap.entrySet()) {
```

```
    System.out.println(entry.getKey() + " : " + entry.getValue());
```

```
}
```

```
// HashSet
```

```
HashSet<String> courses = new HashSet<>();
```

```
courses.add("Java");
```

```
courses.add("Python");
```

```
courses.add("Java"); // Duplicate ignored
```

```
System.out.println("\nHashSet:");
```

```
for (String course : courses) {
```

```
    System.out.println(course);
```

```
}
```

```
// Try-Catch Exception
```

```
try {
```

```
    int result = 10 / 0;
```

```

        System.out.println(result);
    } catch (ArithmeticException e) {
        System.out.println("\nException Caught: " + e.getMessage());
    }

    // Custom Exception Ex

    try{
        checkAge(15);
    } catch (InvalidAgeException e) {
        System.out.println("Custom Exception: " + e.getMessage());
    }
}
}

```

Output:

```

ArrayList:
Isha
Rahul
Aman

HashMap:
101 : Isha
102 : Rahul

HashSet:
Java
Python

Exception Caught: / by zero
Custom Exception: Age must be 18 or above.

```

GitHub Link: <https://github.com/isha05-alt/Tasks>